RACE
Regular Analysis Through Continuous Exercise

General Principles and Processes of Isolation of Elements-Metallurgy: Race-01

1. Mac Arthur process is used for :-
(A) Ag (B) Fe (C) Cl (D) O_2

2. The metal that occurs in the native state as well as in the combined form is :
(A) Silver (B) Magnesium (C) Aluminium (D) Manganese

3. Which of the following is not an example of oxide ore ?
(A) Horn silver (B) Cassiterite (C) Cuprite (D) Haematite

4. Important ore of Zinc is:
(A) Calamine (B) Magnetite (C) Cryolite (D) Anglesite

5. Among the following statements, the incorrect one is:
(A) Calamine and siderite are carbonates (B) Argentite and bauxite are oxides
(C) Zinc blende and galena are sulphides (D) Malachite and azurite are ores of copper

6. Match the ores listed in column I with the type of ores listed in column II and select the correct alternate.

Column I	**Column II**
(1) Bauxite.	(p) Carbonate ore.
(2) Argentite.	(q) Halide ore.
(3) Carnallite	(r) Sulphide ore.
(4) Azurite.	(s) Oxide ore.

Pankers Academy JEE

Codes :

	(1)	(2)	(3)	(4)
(A)	(s)	(r)	(q)	(p)
(B)	(p)	(s)	(q)	(r)
(C)	(p)	(q)	(r)	(s)
(D)	(s)	(r)	(p)	(q)

7. Gravity separation method is based upon:
(A) Preferential washing of ores and gangue particles.
(B) Difference in densities of ore particles and impurities.
(C) Difference in chemical properties of ore particles and impurities.
(D) None of these.

8. The process of removing lighter gangue particles by washing in a current of water is called :
(A) Levigation (B) Liquation (C) Leaching (D) Cupellation

9. NaCN is sometimes added in the froth flotation process as a depressant when ZnS and PbS minerals are expected because :
(A) $Pb(CN)_2$ is precipitated while no effect on ZnS.
(B) ZnS forms soluble complex $Na_2[Zn(CN)_4]$ while PbS forms froth
(C) PbS forms soluble complex $Na_2[Pb(CN)_4]$ while ZnS forms froth.
(D) NaCN is never added in froth floatation process.

For More Join: @IITJEE_Advanced

10. Wolframite is separated from cassiterite by :

(A) Froth floatation method. (B) Column chromatography

(C) Electromagnetic separation method. (D) Electrostatic separation method.

11. Silver is obtained from $Na[Ag(CN)_2]$ by reaction with :

(A) Fe (B) Na (C) Zn (D) Au

12. Process of heating ore in air to remove sulphur is :

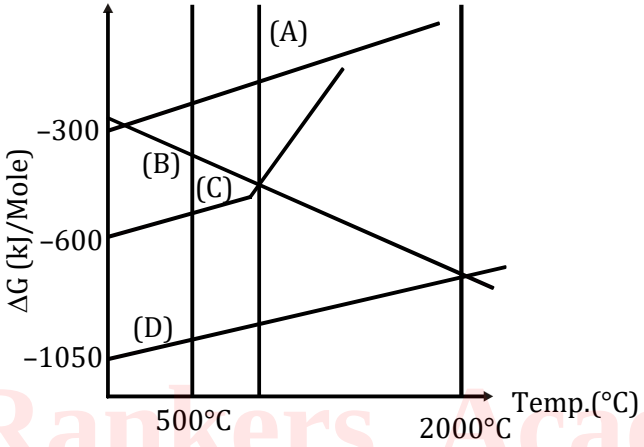
(A) Calcination (B) Roasting (C) Zone refining (D) Liquation

13. Which of the following reaction is an example of calcination process?

(A) $2Ag + 2HCl + [O] \xrightarrow{\Delta} 2AgCl + H_2O$ (B) $2Zn + O_2 \xrightarrow{\Delta} 2ZnO$

(C) $2ZnS + 3O_2 \xrightarrow{\Delta} 2ZnO + 2SO_2$ (D) $MgCO_3 \xrightarrow{\Delta} MgO + CO_2$

14.



Rankers Academy JEE

The element can be used to reduce the oxide of (C) at 2000°C

(A) B only (B) D only (C) A only (D) B & D both

15. Among the following groups of oxides, the group that cannot be reduced by carbon to give the respective metals.

(A) Cu_2O, SnO_2 (B) Fe_2O_3, ZnO (C) CaO, K_2O (D) PbO, FeO

16. Carbon cannot be used to produce magnesium by chemical reduction of MgO because :

(A) Carbon is not a powerful reducing agent

(B) Magnesium reacts with carbon to form carbides

(C) Carbon does not react with magnesium

(D) Carbon is a non-metal

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	A	A	B	A	B	A	B	C
Question	11	12	13	14	15	16				
Answer	C	B	D	D	C	B				

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

General Principles and Processes of Isolation of Elements-Metallurgy: Race-02

1. When lime stone is heated, CO_2 gas is liberated. In metallurgy, this process is called as :
(A) Calcination (B) Roasting (C) Ore dressing (D) Sublimation

2. Self-reduction of Cu_2S to Cu mainly occurs in:
(A) Bessemer convertor (B) Blast furnace
(C) Hall-Heroult tank (D) Both (A) and (B)

3. Which of the following reactions represents the self-reduction process?
(A) $\begin{cases} HgS + O_2 \rightarrow HgO + SO_2 \\ HgO + HgS \rightarrow Hg + SO_2 \end{cases}$ (B) $\begin{cases} Cu_2S + O_2 \rightarrow Cu_2O + SO_2 \\ Cu_2S + Cu_2O \rightarrow Cu + SO_2 \end{cases}$
(C) $\begin{cases} PbS + O_2 \rightarrow PbO + SO_2 \\ PbO + PbS \rightarrow Pb + SO_2 \end{cases}$ (D) All of these

4. Which of the following represents the thermite reaction?
(A) $3Mn_3O_4 + 8Al \xrightarrow{\Delta} 9Mn + 4Al_2O_3$ (B) $2Al(OH)_3 \xrightarrow{\Delta} Al_2O_3 + 3H_2O$
(C) $Cu_2S + 2Cu_2O \xrightarrow{\Delta} 6Cu + SO_2$ (D) $ZnO + C \xrightarrow{\Delta} Zn + CO$

5. Fractional crystallisation method is used for :
(A) The concentration of ore (B) The reduction of metal oxide
(C) The purification of metal (D) The purification of ore

6. The process of zone refining is used for:
(A) Silicon (B) Germanium (C) Gallium (D) All the above.

7. High purity copper metal is obtained by:
(A) Carbon reduction (B) Hydrogen reduction
(C) Electrolytic reduction (D) Thermite reduction

8. The anode mud in the electrolytic refining of silver contains:
(A) Zn, Cu, Ag, Au (B) Zn, Ag, Au (C) Cu, Ag, Au (D) Au only

9. In the electrorefining, the impure metal is made as:
(A) Anode (B) Cathode (C) Electrolytic-tank (D) None

10. Which of the following pairs of metals is purified by van Arkel method?
(A) Ga and In (B) Zr and Ti (C) Ag and Au (D) Ni and Fe

11. Select incorrect statement regarding refining:
(A) Chromatography is useful method for purification of such element which are in minute quantity and impurities are not very different in chemical properties.
(B) Zone refining method is used for purification of semi-conductors.
(C) Liquation method is used for purification of Ti.
(D) Zn and Hg metal are purified by distillation method.

Rankers Academy JEE

For More Join: @IITJEE_Advanced

12. Al is obtained in large quantity by:
(A) Heating cryolite
(B) Reduction of Al_2O_3 with carbon
(C) Reduction of Al_2O_3 with potassium
(D) Electrolytic reduction of Al_2O_3 dissolved in molten cryolite.

13. Hall-Heroult's process is given by :
(A) $2Al_2O_3 + 3C \rightarrow 4Al + 3CO_2$
(B) $Cu^{\oplus 2}(aq) + H_2(g) \rightarrow Cu(s) + 2H^{\oplus}(aq)$
(C) $ZnO + C \xrightarrow{\text{Coke, } 1673K} Zn + CO$
(D) $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$

14. Extraction of zinc from zinc blende is achieved by:
(A) Electrolytic reduction
(B) Roasting followed by reduction with carbon
(C) Roasting followed by reduction with another metal
(D) Roasting followed by self-reduction

15. Refining method which is not used for zinc is -
(A) Liquation
(B) Fractional distillation
(C) Zone refining
(D) Electrolytic refining

16. High purity copper metal is obtained by :
(A) Carbon reduction
(B) Hydrogen reduction
(C) Electrolytic reduction
(D) Thermite reduction

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	D	A	C	D	C	D	A	B
Question	11	12	13	14	15	16				
Answer	C	D	A	B	C	C				

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

General Principles and Processes of Isolation of Elements-Metallurgy : Race-03

1. The following reactions take place in the blast furnace in the preparation of impure iron. Identify the reaction pertaining to the formation of slag.

(A) $CaO(s) + SiO_2(s) \rightarrow CaSiO_3(s)$)

(B) $2C(s) + O_2(s) \rightarrow 2CO(g)$

(C) $Fe_2O_3(s) + 3C(g) \rightarrow 2Fe(I) + 3CO_2(g)$

(D) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

2. Refining of Iron is done by :

(A) Electrolytic method (B) Zone refining

(C) Poling (D) Selective oxidation

3. Bessemer converter is used for :

(A) Pig Iron production (B) Cast iron production

(C) Wrought iron production (D) Steel production

4. Stainless steel contains iron along with :

(A) Cr + Cu + C (B) Ni + Zn + Cr

(C) Cr + Ni + C (D) Ni + Cu + Mn

Bankers Academy JEE

5. Poling process :

(A) Reduces SnO_2 to Sn

(B) Oxidises impurities and remove as scum

(C) Uses green poles

(D) All of the above are correct

6. Tin is extracted from tin stone by :

(A) Self reduction (B) Carbon reduction

(C) Electrolytic reduction (D) Thermite reduction

7. NaCN or KCN is added during froth floatation process of galena, which works as -

(A) Depressant for ZnS impurity (B) Depressant for PbS

(C) Activator for ZnS (D) None of these

8. When impurity content is low then lead is extracted from galena by?

(A) Self reduction (B) Carbon reduction

(C) Electrolytic reduction (D) Hydrometallurgy

9. Direct heating of $MgCl_2.6H_2O$ will produce

(A) $MgCl_2 + H_2O$ (B) $MgO + H_2O$

(C) $MgO + HCl + H_2O$ (D) None of these

For More Join: @IITJEE_Advanced

10. During electrolytic reduction of fused $MgCl_2$, NaCl is mixed to :
(A) Reduce M.P. and increase Conductivity of $MgCl_2$
(B) Increase M.P. and reduce Conductivity of $MgCl_2$
(C) Remove impurities from $MgCl_2$
(D) None of the above

11. Leaching of argentite is done in presence of air :
(A) To oxidise Ag_2S
(B) To prevent formation of Ag_2O
(C) To oxidise Na_2S into Na_2SO_4
(D) To reduce melting point of Ag_2S

12. Zn is used to obtain Ag from $Na[Ag(CN)_2]$, because :
(A) Zn has tendency to gain electron
(B) Standard reduction Potential of Zn is more than Ag
(C) Zn is more electropositive than Ag
(D) None of these

13. Which of the following process in not used in refining of gold ?
(A) Cupellation (B) Parting
(C) Electrorefining (D) Poling

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	D	D	C	D	B	A	A	C	A
Question	11	12	13							
Answer	C	C	D							

For More Join: @IITJEE_Advanced